

3-2 Kinetic Energy and the Work–Kinetic Energy Theorem



Source / 來源

- 選修物理(二) 課堂講義 (4624R51) 教用版，第3章 功與能量，§ 3-2 動能與功能定理，pp. 116–123
- Halliday, Resnick & Walker, *Fundamentals of Physics*, Ch 7: Kinetic Energy and Work
- Young & Freedman, *University Physics with Modern Physics*, Ch 6: Work and Kinetic Energy

Examples

Example 1 — Change in Momentum and Kinetic Energy (*p. 117*)

A 2 kg object moves on a smooth horizontal surface. Its speed is 2 m/s at $t = 0$ and 22 m/s at $t = 2$ s. Find:

(a) The magnitude of its change in momentum.

(b) Its change in kinetic energy.

(c) Does $\Delta E_k = (\Delta p)^2 / (2m)$ hold here?



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Solution

Try It 1 — Force Varying with Time (*p. 117*)

A 2 kg object starts from rest on a smooth surface. A horizontal force $F(t) = 3t + 2$ N acts on it. What is its kinetic energy at $t = 2$ s?

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 Solution

Example 2 — Work–Kinetic Energy Theorem (*p. 118*)

A block starts from rest on a fixed incline of angle θ . A horizontal constant force F pushes it up the incline through distance ℓ ; its speed then is v . Find the work done by kinetic friction. The block has mass m and gravity is g .

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 Solution

Try It 2 — Sliding to Rest (*p. 119*)

An object moves at speed v on a horizontal surface. After traveling distance S , its speed is $v/3$. The resistive force is constant. What total distance does it travel before stopping?

(A) $\frac{4}{3}S$

(B) $\frac{5}{4}S$

(C) $\frac{7}{6}S$

(D) $\frac{9}{8}S$

(E) $\frac{8}{7}S$

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 Solution

Quick Check

Kinetic Energy

Q1. Two particles have masses m_1, m_2 , momenta of equal magnitude $p_1 = p_2$, and $m_1 > m_2$. Which relation is correct?

(A) $m_1 E_{k1} > m_2 E_{k2}$

(B) $m_1 E_{k1} = m_2 E_{k2}$

(C) $m_1 E_{k1} < m_2 E_{k2}$

(D) $m_1 E_{k2} = m_2 E_{k1}$

(E) $m_1 E_{k2} > m_2 E_{k1}$

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 Solution

Q2. An object of mass m initially has kinetic energy T . Its speed increases by Δv . Find ΔE_k .

(A) $2mT\Delta v$

(B) $\frac{1}{2}m(\Delta v)^2$

(C) $2mT\Delta v + \frac{1}{2}m(\Delta v)^2$

(D) $T + \frac{1}{2}m(\Delta v)^2$

(E) $mT\Delta v + \frac{1}{2}m(\Delta v)^2$

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 Solution

Q3. A 10 kg object slides down a smooth incline from rest, then moves 5 m on a horizontal surface with $\mu_k = 0.4$ before stopping. What was its original vertical height?

(A) 0.4 m

(B) 1 m

(C) 2 m

(D) 5 m

(E) 9.8 m

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 Solution

Q4. A particle slides from rest down a smooth incline of length L . Its kinetic energies at distances $L/3$ and L are E_1 and E_0 , respectively. Find E_1/E_0 .

(A) 1/9 (B) 1/3 (C) 4/9 (D) 2/3 (E) 1

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 Solution

Q5. A 2 kg block starts from rest on a smooth horizontal surface. The $F-t$ graph is a triangle of base 3 s and height 20 N. How much work does the force do in 3 s?

(A) 30 J (B) 125 J (C) 225 J (D) 250 J (E) 375 J

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 Solution

Challenge

Kinetic Energy

Q1. A planet of mass m orbits the Sun in an ellipse. Its greatest and least distances are R and r , and its minimum speed is v . What is the kinetic energy at perihelion minus that at aphelion?

(A) 0

(B) $\frac{m(R-r)v^2}{2r}$

(C) $\frac{m(r-R)v^2}{2R}$

(D) $\frac{m(R^2 - r^2)v^2}{2r^2}$

(E) $\frac{m(r^2 - R^2)v^2}{2R^2}$

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 Solution

Work-Kinetic Energy Theorem

Q2. On a smooth horizontal surface, stationary objects of masses m and $2m$ are acted on by the same horizontal force F through the same distance S . Which statement is correct?

- (A) F does more work on the $2m$ object.
- (B) F does more work on the m object.
- (C) The two objects gain equal kinetic energy.
- (D) The two objects gain equal momentum.
- (E) The two objects have equal average power.

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 Solution

Q3. A person of mass $2m$ stands on a cart of mass m moving at speed V on a frictionless track. The person jumps off and has zero velocity relative to the ground. What work does the person do on the cart?

(A) $\frac{1}{3}mV^2$ (B) mV^2 (C) $2mV^2$ (D) $3mV^2$ (E) $4mV^2$

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 Solution

Q4. A bullet of initial speed v passes through a fixed block of thickness d and leaves at $v/2$. If a thickness d' just stops the bullet, find d' .

(A) $4d/3$ (B) $2d$ (C) $3d/2$ (D) $5d/3$ (E) $2d/5$

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 Solution

Q5. A ball of initial speed v_0 moves on a smooth horizontal surface. A constant force F makes an angle 45° with the initial velocity and acts for time Δt . Find the work done by F .

(A) $Fv_0\Delta t$ (B) $F(\Delta t)^2/(2m)$ (C) $\frac{Fv_0\Delta t}{\sqrt{2}} + \frac{F^2(\Delta t)^2}{2m}$ (D) $\frac{2Fv_0\Delta t}{2} + \frac{F^2(\Delta t)^2}{2m}$ (E) $Fv_0\Delta t + \frac{F^2(\Delta t)^2}{2m}$

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 Solution

Conveyor-Belt Group

A horizontal conveyor belt moves at constant speed v . A box of mass m is placed on it at $t = 0$ with zero horizontal speed. The kinetic friction coefficient is μ_k .

Q6. What work does friction do while the box is accelerating to the belt speed?

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 Solution

Q7. After what time does friction cease to do work on the box?

 Solution

Exam Archive

Q1 — Incline with an Applied Force (*p. 123, 108 指考*)

A mass m starts at rest at the bottom P of a fixed 60° rough incline of length L . A constant force F acts up the incline until the mass reaches midpoint Q , where the force is removed. The coefficient of kinetic friction is μ . What minimum F allows the mass to reach the top R ?

- (A) $\frac{3}{2}\mu mg$ (B) $(\frac{3}{2} + 1)\mu mg$ (C) $\frac{\mu + \sqrt{3}}{2}mg$ (D) $(\frac{3\mu + 1}{2})mg$ (E) $(\mu + 3)mg$

 Solution

Q2 — Work from an Acceleration–Time Graph (*p. 123, 103 指考*)

A 2 kg object starts from rest on a smooth surface. The signed area under its acceleration–time graph from $t = 0$ to 4 s is 2 m/s. Find the work done by the external force.

- (A) 0 (B) 2 J (C) 4 J (D) 6 J (E) 8 J

 Solution

Q3 — Work from an F - x Graph (*p. 123, 102 指考*)

The force on a particle varies with position as shown in an F - x graph. The particle starts at $x = 0$ moving in the $+x$ direction. At which labeled position is its speed greatest?

 Solution

Q4 — Kinetic-Friction Work (*p. 123, 92 指考*)

A 2.0 kg block is pushed horizontally by a constant 6.0 N force on a table with $\mu_k = 0.25$. How much does its kinetic energy increase after moving 5.0 m? Take $g = 10 \text{ m/s}^2$.

(A) 30 J (B) 15 J (C) 10 J (D) 5 J (E) 2 J

Glossary

English	中文	Symbol / Unit
kinetic energy	動能	$E_k = \frac{1}{2}mv^2$ [J]
momentum	動量	$p = mv$ [kg · m/s]
impulse	衝量	$J = \Delta p$ [N · s]
change in kinetic energy	動能變化量	ΔE_k
work–kinetic energy theorem	功能定理	$W_{\text{net}} = \Delta E_k$
net work	合力作功	W_{net}
external force	外力	F_{ext}
smooth surface	光滑面	—
rough surface	粗糙面	
kinetic friction	動摩擦力	$f_k = \mu_k N$
acceleration–time graph	加速度–時間圖	$a-t$
force–position graph	力–位置圖	$F-x$
Kepler's second law	克卜勒行星運動第二定律	equal areas in equal times